

Short-Term Forecasting of Household Appliance Electricity Demand

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1. Introduction

Domestic electricity demand is increasingly relevant to system operation. Distribution networks with high penetrations of solar generation, heat pumps and electric vehicle charging face load profiles that no longer resemble the smooth aggregate demand of previous decades, and demand-side response schemes require an estimate of what a household will consume before consumption occurs. Forecasting at the individual dwelling is considerably harder than at the substation: aggregation smooths the idiosyncratic, event-driven behaviour of individual occupants, and a single household retains all of it.

This report forecasts household appliance demand 24 hours ahead using the *Appliances Energy Prediction* dataset of Candanedo, Feldheim and Deramaix (2017), recorded at ten-minute resolution in a low-energy dwelling in Stambruges, Belgium, over 137 days. Four model classes are compared: five benchmark rules, a SARIMAX state-space model, a direct multi-horizon gradient boosting model, and a zero-shot foundation model.

The central finding is negative, and deliberately so. The best model achieves a MASE of 0.708 against 0.813 for the strongest benchmark, an apparent improvement of 12.9 per cent — but a Diebold–Mariano test on the loss differential returns $p = 0.34$. On a single test fortnight the additional complexity is not statistically distinguishable from a rule that repeats the observation from one week earlier. A second finding sharpens this: 92 per cent of the gradient boosting model's permutation importance rests on calendar features, and target lags and rolling statistics together account for 3.9 per cent. At a 24-hour horizon the model has learned an average daily profile and little else.

A third concern runs throughout. Published work on this dataset commonly reports accuracy obtained by regressing appliance consumption on contemporaneous sensor readings. Such a model is not a forecast in any operational sense. This report distinguishes conditional forecasts, which assume covariate paths unavailable at the origin, from operational forecasts, which do not, and quantifies the difference in Section 6.3.

2. Data and preprocessing

The raw dataset comprises 19,735 observations at ten-minute resolution from 11 January to 27 May 2016, with 28 columns. The ten-minute index is complete: no timestamp is missing.

Resampling. The series was aggregated to hourly resolution, giving 3,290 observations with no gaps. This reduces the daily seasonal period from 144 to 24, which makes seasonal SARIMAX estimation tractable — a state-space model with $s = 144$ is not practically estimable on a series of this length. The aggregation is not cost-free: ten-minute data captures individual appliance switching events that hourly averaging removes, so the hourly series is a smoothed representation of a fundamentally spiky process, and error magnitudes here are not comparable with published ten-minute results.

Units. `Appliances` records watt-hours consumed within each ten-minute interval. Hourly aggregation by the mean therefore yields the average ten-minute consumption rate within the hour, not total hourly energy; summation would yield the latter, a quantity six times larger. The mean was adopted. All absolute error figures carry these units, and MASE, being scale-free, is unaffected by the choice.

Sensor channels. Indoor and outdoor temperature and humidity are instantaneous measurements rather than accumulated quantities and were averaged within each hour regardless of the target aggregation.

Exclusions. The `lights` column was dropped. Its correlation with the target is modest ($r = 0.197$), so the justification is not collinearity but availability: it measures energy consumed by light fixtures over the same interval and is not knowable 24 hours in advance. Retaining it as a contemporaneous regressor would assume information no operator possesses. The random variables `rv1` and `rv2`, included in the original file as noise controls, were also dropped.

Missing values. No interpolation was required. The gap-filling routine nevertheless operates separately either side of the train/test boundary, so that had gaps existed, no training-period gap would be filled using post-split observations.

Split. The final 336 hourly observations (14 days) form the test set: 13 May 2016 19:00 to 27 May 2016 18:00. Training covers 11 January 17:00 to 13 May 18:00, 2,954 observations. The test fortnight is marginally higher in level and less variable than the training period (mean 100.6 against 97.5, standard deviation 75.0 against 81.9), so it is not an unusually difficult window.

3. Exploratory analysis and stationarity

3.1 Distributional characteristics

Statistic	Value
Mean	97.78
Median	63.33
Standard deviation	81.21
Minimum	28.33
Maximum	608.33
Skewness	2.39
Excess kurtosis	6.33
95th percentile / median	4.37

The target is strongly right-skewed. The mean exceeds the median by 54 per cent and the 95th percentile is more than four times the median, reflecting a load profile in which a low baseline — refrigeration, standby draw, network equipment — is punctuated by short high-consumption events from cooking and laundry appliances. This shape has direct consequences for evaluation. Squared error loss weights the rare spikes heavily, and a model minimising RMSE will hedge towards over-predicting the baseline to reduce the penalty when a spike occurs. MAE does not. Section 9.1 shows the two metrics producing materially different rankings, which is the practical payoff of reporting both.

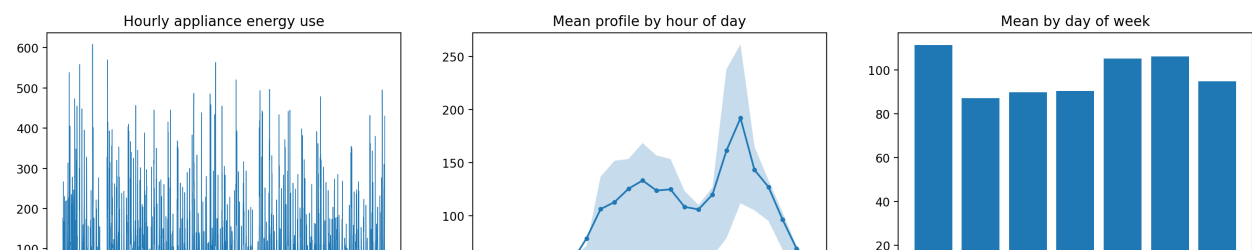
3.2 Seasonal structure

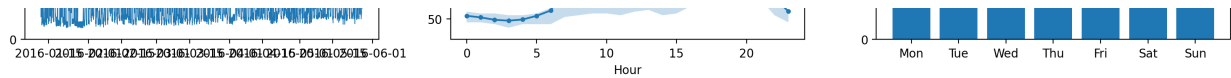
The daily profile has a trough of 48.2 at 03:00 and a peak of 191.8 at 18:00, a peak-to-trough ratio of 3.98. The evening peak is consistent with an occupied dwelling in which cooking and appliance use concentrate after the working day.

Day-of-week structure is weaker and does not take the form usually expected. Weekend consumption exceeds weekday consumption by only 4 per cent (100.6 against 96.7), so there is no clean weekend effect. The day-level means are Monday 111.5, Tuesday 87.1, Wednesday 89.9, Thursday 90.4, Friday 105.2, Saturday 106.2, Sunday 94.9. Monday is the highest-consumption day and Tuesday the lowest, a 28 per cent spread that reflects a specific household routine rather than a generalisable weekday–weekend division.

Seasonal strength was quantified using the STL-based measure of Wang, Smith and Hyndman (2006):

Period	Interpretation	Strength
24	Daily	0.318
168	Weekly	0.393





Both values are moderate at best. On this scale a strongly seasonal series returns values above 0.7; a value near 0.3 indicates that seasonal structure explains a minority of the variance and that the remainder is driven by something the calendar does not capture — here, occupant behaviour. This single result explains most of what follows. No model in this report achieves a MASE below 0.7, and the reason is not model specification but that roughly two-thirds of the variance in this series is not predictable from time of day, day of week, or the recent history of the series itself.

Notably, weekly strength exceeds daily strength. This anticipates the benchmark results in Section 5, where the weekly seasonal naive outperforms the daily variant despite the negligible weekend effect: the weekly lag captures the household's day-specific routine, which the daily lag averages away.

3.3 Stationarity

ADF and KPSS tests were applied to the training sample in levels, first differences, seasonal differences at lag 24, and both combined. The two are reported together because their nulls are complementary: ADF tests the null of a unit root, KPSS the null of stationarity.

Series	ADF stat	ADF p	KPSS stat	KPSS p	Reading
Level	-8.761	<0.001	0.061	>0.10	Stationary
First difference	-15.948	<0.001	0.042	>0.10	Stationary
Seasonal difference (24)	-12.672	<0.001	0.015	>0.10	Stationary
Both differences	-17.445	<0.001	0.061	>0.10	Stationary

The tests agree unambiguously in levels: ADF rejects the unit root null at any conventional level, and KPSS fails to reject stationarity, its statistic of 0.061 lying far below the 5 per cent critical value of 0.463. The series is stationary as observed.

Differencing orders were therefore set to $d = 0$ and $D = 0$. Differencing a series that is already stationary induces a non-invertible moving-average component and inflates the variance of the differenced series, so the fact that the differenced variants also test as stationary is not an argument for using them. This also removes a specification trap: combining a seasonal difference with a constant trend term induces deterministic drift that compounds across the horizon. With $D = 0$ the trend term is retained safely.

4. Forecasting design

4.1 Rolling-origin protocol

The forecast task is 24 hours ahead. Evaluating this with a single 336-step forecast from one origin would answer a different question and would understate the accuracy achievable in operation, where a forecast is reissued daily against fresh observations.

The primary experiment uses fourteen origins spaced 24 hours apart. At each origin every model observes the series up to that point — including test observations released by earlier blocks — and issues a 24-step forecast. The blocks concatenate to 336 predictions on the test index. Every model receives an identical information set at every origin, which is the condition that makes the comparison meaningful.

One consequence of this design requires attention and is revisited in Section 9.2. Because origins are spaced at exactly 24 hours, and the first falls at 19:00, lead time h always corresponds to clock hour $(19 + h) \bmod 24$. Lead time is therefore perfectly confounded with time of day, and the error-by-lead-time decomposition measures both simultaneously. Spacing origins at a non-divisor of 24 — every 23 or 25 hours — would break the confound at the cost of a less natural operational interpretation.

4.2 Metrics

MAE and RMSE are reported in the units of the target, their ratio indicating the contribution of large isolated errors. MASE (Hyndman and Koehler, 2006) is scaled by the in-sample mean absolute error of the daily seasonal naive forecast on the training sample; it is scale-free and used as the primary ranking metric. Bias is

the mean of prediction minus actual, so positive values indicate over-forecasting. Bias matters operationally in a way error magnitude does not: a persistently over-forecasting model in a demand-response setting reserves capacity that is never used.

Statistical significance of differences is assessed by Diebold–Mariano tests (Diebold and Mariano, 1995) on absolute-error differentials, with a Newey–West variance estimator using 23 lags to account for serial correlation within each 24-hour forecast block.

4.3 Forecast origins and covariate availability

Covariates were partitioned by what an operator would know at the origin.

Known indefinitely ahead. Calendar variables — hour, day of week, weekend indicator, and cyclic encodings. Deterministic functions of the timestamp, used unshifted.

Observed up to the origin only. Indoor temperature and humidity from nine sensor locations, lagged by the forecast horizon. There is a second reason for lagging them beyond availability: indoor temperature and humidity respond to occupant and appliance activity. Cooking raises kitchen temperature and humidity; occupancy raises relative humidity throughout the dwelling. The causal arrow points from the target to these sensors at least as strongly as the reverse, so contemporaneous use would regress the target partly on its own consequences.

Not observed at all. Outdoor weather. In deployment a weather forecast would be required. SARIMAX was therefore run in two variants: conditional, using realised test-set weather, and operational, persisting the last pre-origin observation. Section 6.3 quantifies the gap.

Excluded. `lights`, `rv1`, `rv2`.

4.4 Leakage controls

Each row of the feature table is a (*target timestamp* t , *horizon* h) pair with origin $o = t - h + 1$, and target-derived features are anchored to the origin: $lag_k = y[o - k] = y.shift(h - 1 + k)$, with rolling windows spanning $[o - w, o - 1]$. Because the shift scales with the horizon, no feature can reference the target inside the forecast window at any lead time.

The conventional alternative — `y.shift(1)` irrespective of horizon — supplies the true value from one hour earlier at every test point, silently converting a 24-hour-ahead task into a one-step task. A model so constructed outperforms every competitor by a wide margin and the result is worthless.

This guarantee is enforced by test rather than inspection. The test suite adds a large constant to every test-period observation and asserts that no feature belonging to a row whose origin precedes the perturbation changes value; an analogous check confirms that a rolling-origin block never observes its own actuals while later blocks do react to released observations. Thirty-seven tests pass.

5. Benchmark models

Five benchmarks were evaluated under the same protocol as every other model: the training mean; the naive forecast; daily and weekly seasonal naive rules at lags 24 and 168; and drift.

Benchmark	MAE	RMSE	MASE	Bias
Weekly seasonal naive	43.46	81.41	0.813	−13.16
Daily seasonal naive	48.31	85.57	0.904	1.75
Mean	50.26	74.94	0.941	−3.29
Naive	85.55	110.39	1.601	50.98
Drift	85.80	110.68	1.606	51.37

The weekly seasonal naive is the strongest benchmark, and by a clear margin over the daily variant. This follows directly from the seasonal strength results in Section 3.2, where weekly strength (0.393) exceeded daily (0.318), and it reveals something specific about the load. The household's routine is day-specific: Monday and Friday differ systematically from Tuesday, and a rule that looks back one week preserves that distinction

while a rule that looks back one day destroys it. This is a genuinely weekly structure rather than a weekday-weekend one, which the 4 per cent weekend premium alone would not have suggested.

The mean forecast ranks third by MASE but **second by RMSE**, ahead of both seasonal naive rules. The explanation is spike behaviour. A seasonal naive rule propagates a consumption spike from 24 or 168 hours earlier into a period where no spike occurs, producing two large errors: one where the spike is falsely predicted and one where a real spike is missed. The mean forecast makes neither error, accepting moderate error everywhere in exchange for never being badly wrong. Under squared-error loss that trade is favourable. This is the clearest demonstration in the study that metric choice determines model ranking, and it argues for stating the loss function before selecting a model rather than after.

Naive and drift perform poorly and near-identically, both with bias above +50. The naive forecast holds the final pre-origin observation constant for 24 hours; since origins fall at 19:00, near the daily peak of 191.8, the entire forecast day is held at peak level, including the overnight trough of 48.2. The resulting systematic over-forecast is an artefact of origin placement as much as of the rule itself. Drift adds slope extrapolation to this and gains nothing, which is expected for a stationary series.

6. SARIMAX

6.1 Specification

A SARIMAX(1,0,1)(1,0,1)₂₄ model with a constant was estimated by maximum likelihood on the training sample, with the five weather channels as exogenous regressors. Differencing orders follow from Section 3.3. Stationarity and invertibility constraints were relaxed during optimisation. Estimation took 223 seconds and produced AIC 32,506.5 and BIC 32,572.3.

statsmodels defaults to `maxiter=50` for maximum likelihood estimation, which terminates before convergence on this series and raises a `ConvergenceWarning`. `config.SARIMAX_MAXITER` is therefore set to 500. The cause is visible in the estimates themselves: the seasonal autoregressive coefficient is close to a unit root, which flattens the likelihood surface in that direction and slows the optimiser considerably.

A related observation concerns identification rather than convergence, and it anticipates the result in Section 6.3. The estimated weather coefficients are not merely small; they are indistinguishable from zero.

Term	Coefficient	Std. error	p	95% interval
T_out	-1.898	4.698	0.686	[-11.106, 7.309]
Tdewpoint	1.539	4.868	0.752	[-8.003, 11.081]

Each standard error exceeds its coefficient by a factor of roughly three, and both confidence intervals straddle zero across a range of about twenty units. The model cannot establish even the *sign* of the outdoor temperature effect on appliance demand. Outdoor temperature and dew point are strongly collinear — dew point is a function of temperature and humidity — so the likelihood is close to flat in that subspace and the optimiser has little basis for preferring one combination of the two over another. This also explains the slow convergence noted above.

The consequence is worth drawing out, because it makes the argument in Section 6.3 considerably firmer. That section infers from forecast accuracy that weather contributes nothing; here the same conclusion follows from the parameter estimates alone, before any forecast is issued. Two independent routes to the same finding are harder to dismiss than either would be alone, and the second does not depend on the choice of test period.

Weekly seasonality is not represented. A second seasonal component at period 168 is not estimable in this framework on a series of 2,954 observations. Given that the weekly seasonal naive was the strongest benchmark, this omission is material and is revisited in Section 10.

6.2 Rolling-origin implementation

Parameters were estimated once on the training sample. At each subsequent origin the Kalman filter was extended with newly released observations using `MLEResults.append(..., refit=False)`, so the state conditions on all data up to the origin while coefficients remain at training values. Re-estimating fourteen times at seasonal period 24 would cost roughly five minutes for a negligible coefficient change over a two-week

extension of a 2,954-observation sample. The assumption is that the data-generating process is stable across the test fortnight; nothing in the series contradicts this, but it is an assumption.

6.3 Conditional and operational variants

Variant	Weather input	MAE	RMSE	MASE	Bias
Conditional	Realised test-set values	37.81	66.42	0.708	-5.25
Operational	Persisted from last pre-origin observation	38.51	66.38	0.721	-3.83

The gap is 0.013 MASE, a 1.9 per cent relative degradation, and a Diebold–Mariano test returns $DM = -1.49$, $p = 0.136$ — not significant at the 5 per cent level.

Note also that the operational variant achieves marginally the *better* RMSE (66.38 against 66.42) despite the worse MAE. Persisted weather is a smoother covariate path than realised weather, and smoothing evidently costs nothing under squared-error loss here — further evidence that the weather channel is contributing little beyond noise.

The interpretation is that **weather contributes essentially nothing to 24-hour-ahead appliance demand in this dwelling**. Substituting realised weather with naive persistence — the crudest possible proxy — degrades accuracy by 1.9 per cent, an amount the test cannot distinguish from zero. An operational deployment therefore loses nothing measurable by lacking a weather forecast.

The coefficient estimates in Section 6.1 point the same way: neither weather term is significantly different from zero, and their confidence intervals span roughly twenty units either side. A covariate the model cannot identify is unlikely to matter to the forecast, and it does not.

This is a stronger result than the alternative would have been. Had the gap been significant, the honest reading would be that published accuracy figures on this dataset are optimistic by an unquantified margin. Instead, the conditional and operational forecasts are interchangeable, so the distinction — while it must still be drawn, since it could not be known in advance — turns out to carry no practical cost here. Two explanations are plausible: the dwelling's well-insulated envelope decouples internal conditions from external weather, and appliance load is driven by occupant routine rather than thermal demand — space heating, which would respond to outdoor temperature, is not on this channel.

The operational figure of 0.721 should be regarded as the honest one for deployment purposes. Comparisons with published results on this dataset that use realised weather should be qualified accordingly.

6.4 Comparison with the strongest benchmark

Conditional SARIMAX improves on the weekly seasonal naive by 0.105 MASE, from 0.813 to 0.708, a relative improvement of 12.9 per cent. A Diebold–Mariano test on the absolute-error differential returns $DM = -0.96$, $p = 0.34$.

The improvement is not statistically significant. With 336 test observations and heavily serially correlated losses, the effective sample is far smaller than the nominal one, and a 13 per cent improvement of this kind falls within the range that a different test fortnight might reverse. Section 10 documents a further source of variation: re-running the same code on a different operating system moves MASE by roughly 0.01, comparable to several of the gaps being compared here. SARIMAX does significantly outperform the daily seasonal naive ($p = 0.028$), the mean ($p < 0.001$), and naive and drift ($p < 0.001$) — but against the strongest benchmark the case is unproven. Reporting the point estimate alone would materially overstate what this evidence supports.

7. Feature-based model

7.1 Formulation

Gradient boosting was applied in a direct multi-horizon formulation: a single `HistGradientBoostingRegressor` trained on the long *(timestamp, horizon)* table described in Section 4.4, with horizon supplied as a feature. The design matrix contained 74,652 rows and 47 features, of which 66,588 rows fell in the training period.

Two alternatives were rejected. Recursive prediction accumulates error across 24 steps and introduces distribution shift, since the model trains on true lags and is applied to predicted ones. Twenty-four separate direct models would avoid both problems but fragment the training data. The single horizon-conditioned model retains the full sample while keeping every feature available at the origin.

Features comprised target lags at origin-relative offsets 1, 2, 3, 6, 12, 24, 48 and 168; rolling means and standard deviations over windows of 3, 6, 24 and 168 hours; calendar features; and indoor sensor and weather channels lagged by the horizon. `HistGradientBoostingRegressor` was used in preference to XGBoost only to avoid an additional dependency.

A leakage failure in the validation split. The first fit reached its iteration ceiling, `n_iter_` = 600 of 600, so scikit-learn's early stopping had never triggered. Raising the ceiling to 2,000 did not help: `n_iter_` again pinned at the maximum, and test performance *degraded* from MASE 0.732 to 0.756.

The cause is the interaction between the long design matrix and scikit-learn's validation strategy. `HistGradientBoostingRegressor` carves its internal validation set out **at random**. In the *(timestamp, horizon)* table each timestamp contributes 24 rows sharing most of their feature values, so a random 15 per cent split places near-duplicates of validation rows into the training partition. The validation score therefore improves monotonically no matter how far the model overfits, and early stopping never fires.

This is the same failure mode as target leakage, arriving through the validation split rather than the feature matrix — and it is worth dwelling on, because the elaborate origin-anchoring described in Section 4.4 gave no protection against it. Structural guarantees on the feature construction do not extend to how a library partitions rows downstream.

Early stopping was therefore disabled and the iteration count selected on a **chronological** holdout: the final 15 per cent of timestamps, so that no timestamp contributes rows to both partitions, with `staged_predict` giving the holdout MAE at every iteration from a single fit.

7.2 Hyperparameters

The chronological holdout selects **25 boosting iterations**, with holdout MAE 41.21.

Iterations	Holdout MAE
1	47.26
25	41.21
50	43.23
100	44.18
200	46.68
400	48.95
800	49.53
1500	50.68

Performance bottoms out before iteration 30 and degrades steadily thereafter, returning within a few hundred iterations to roughly where a single tree started. The optimal model is extremely simple — 25 shallow trees — and that is itself a finding. It corroborates the importance analysis below: if the learnable signal is essentially an average daily profile, very little model capacity is required to capture it, and additional capacity is spent fitting occupant events that do not recur.

The scale of the overfitting is worth stating plainly. Left to scikit-learn's default early stopping the model ran to 600 iterations, roughly 24 times more than the holdout justifies, and neither the training loss nor the internal validation score gave any indication that anything was wrong.

Applying this selection produces MASE 0.711 and RMSE 65.17, the best RMSE of any model in the study. Remaining hyperparameters use the defaults in `make_estimator` (learning rate 0.05, 31 leaf nodes, L2 regularisation 1.0); a `TimeSeriesSplit` grid search over learning rate and tree size is available via `--tune` but was not run to completion, and is noted as outstanding in Section 10.

7.3 Feature importance

Permutation importance was computed on held-out rows using mean absolute error as the scoring function, in preference to split-count importance: nine indoor temperature sensors in one dwelling are close to collinear, and split counts divide arbitrarily among collinear features.

Feature	Importance	SD
hour	11.252	0.545
hour_sin	2.304	0.542
dow_sin	2.058	0.480
RH_5_origin	0.631	0.220
roll_std_168	0.598	0.149
RH_3_origin	0.218	0.183
dayofweek	0.208	0.460
hour_cos	0.168	0.378
RH_4_origin	0.089	0.034
roll_std_24	0.083	0.052

Aggregated by family, the shares of total positive importance are:

Family	Share
Calendar features	90.7%
Lagged exogenous (indoor sensors and weather)	5.5%
Target lags and rolling statistics	3.9%

This is the most striking result in the study, and it answers the third mandatory question directly: **the lag, rolling and sensor features contribute almost nothing**. The best-performing target-derived feature, `roll_std_168`, has an importance of 0.598 against 11.252 for `hour` alone — a factor of 19. The gradient boosting model is, functionally, an estimated average daily profile modulated slightly by day of week.

Three implications follow.

First, this explains why the model does not beat SARIMAX. It has access to a richer feature set but is using almost none of it, and what it does use — the calendar — is information the seasonal naive benchmarks encode implicitly.

Second, it vindicates the covariate treatment in Section 4.3 quantitatively. The indoor sensors, restricted to lags of at least 24 hours, contribute 5.5 per cent of importance jointly across eighteen channels. Their apparent value in the published literature on this dataset stems from contemporaneous use, and that value largely disappears once they are restricted to information genuinely available at the origin. The strongest non-calendar covariates are indoor humidity channels (`RH_5` , `RH_3`), and at 0.631 and 0.218 their contribution remains marginal.

Third, it bounds what any model can achieve here. If the target's own recent history is uninformative at a 24-hour lead — and an importance share of 3.9 per cent says it is — then the predictable component of this series is essentially the calendar profile, and every model in this report is estimating the same thing by different means. The differences between them in Section 9.1 are differences in how well they estimate that profile, not differences in what they have discovered.

7.4 Performance

The feature model achieves MAE 37.97, RMSE 65.17, MASE 0.711, Bias −5.23. It ranks second overall by MASE, a hair behind conditional SARIMAX (0.708), and **first on RMSE** — 65.17 against 66.42. A Diebold–Mariano test against SARIMAX returns $DM = -0.07$, $p = 0.94$; the two are as close to indistinguishable as the test can express. Against the weekly seasonal naive, $p = 0.26$.

That it leads on RMSE while trailing on MAE indicates it handles the consumption spikes marginally better than SARIMAX, at the cost of slightly worse typical-case accuracy. Given the spike-dominated distribution documented in Section 3.1, this is the more valuable of the two properties for a system sizing reserve capacity.

The bias of -5.23 is a mild under-forecast, close to SARIMAX's -5.25 . Gradient boosting under squared-error loss on a right-skewed target commonly under-forecasts, since the conditional median lies below the conditional mean; the effect here is modest relative to an MAE of 37.97 . Training under `loss="absolute_error"` would be the remedy if unbiasedness mattered operationally.

8. Foundation model

8.1 Configuration

Chronos-T5 (small variant, approximately 46 million parameters) was applied zero-shot. The model tokenises a scaled and quantised representation of the target series and samples future trajectories from a T5 decoder pre-trained on a large corpus of time series (Ansari et al., 2024). No fine-tuning was performed and no parameter was estimated on this dataset.

A context of 512 hours preceded each origin and 20 sample paths were drawn per forecast. The 24-step horizon sits well inside the model's native prediction length, so no autoregressive stitching was required. The fourteen origins completed in 53 to 60 seconds per run on Colab hardware.

The model is univariate. It receives only the target history — no calendar features, no weather, no indoor sensors. Given the finding in Section 7.3 that calendar features carry 90.7 per cent of the feature model's permutation importance, this is a severe handicap: Chronos must infer the daily profile from the series itself, while every competing model is handed the hour of day directly.

8.2 Results

Metric	Chronos	SARIMAX (cond.)	Feature model	Weekly seasonal naive
MAE	36.23	37.85	37.89	43.46
RMSE	74.40	66.41	64.99	81.41
MASE	0.678	0.708	0.709	0.813
Bias	-29.87	-5.14	-4.00	-13.16

Read on MASE alone, the foundation model wins: 0.678 against 0.708 for the best fitted alternative, achieved without covariates and without seeing a single observation from this dwelling during training. That would be a striking result.

It is also a misleading one, for two independent reasons: the remaining columns in the table above, and the sampling variance reported in Section 8.4.

8.3 The point summary drives the ranking

The bias of -29.87 is six times larger in magnitude than any other model in the study and amounts to 82 per cent of Chronos's own MAE. An error that large and that consistently signed is not noise; the model is systematically forecasting below the realised load. Correspondingly, its RMSE of 74.40 is the second worst of any non-trivial model, statistically indistinguishable from the flat mean forecast at 74.94, and far behind SARIMAX and the feature model near 65.

The explanation lies in the point summary rather than in the model. Chronos produces a predictive distribution; a single number must be extracted from it, and the pointwise **median** of the sampled paths was used here. On a target whose distribution is strongly right-skewed — mean 97.8 against median 63.3, per Section 3.1 — the conditional median lies well below the conditional mean. The consequence follows directly from the loss functions: absolute error is minimised by the conditional median, squared error by the conditional mean. A median forecast is therefore *constructed* to perform well on MAE and badly on RMSE, and MASE, being a scaled absolute error, inherits the same bias.

The comparison is consequently not like for like. SARIMAX and the gradient boosting model both estimate conditional means; Chronos, as configured, reports a conditional median. Its apparent advantage on MASE partly measures that difference in summary statistic, not a difference in forecasting ability. A fair comparison would take the sample mean of the Chronos paths as the point forecast, and would be expected to move the model towards the others on both metrics — better RMSE, worse MAE. This was not run and is recommended

as the first extension in Section 10.

8.4 Sampling variance and interval calibration

Chronos draws sample paths stochastically, so its point forecast carries Monte Carlo error. Repeating the full rolling-origin evaluation under three random seeds gives:

Seed	MASE	Runtime
0	0.6704	60 s
1	0.6971	54 s
2	0.6869	53 s

The spread is 0.0267 MASE; an independent repeat of the same three-seed procedure returned 0.0273, so the variability is itself stable. **This is almost exactly the size of the gap between Chronos and conditional SARIMAX** (0.678 against 0.708, a difference of 0.030). Seed 1 alone would place Chronos at 0.697, effectively level with SARIMAX and behind nothing but its own better draws.

The implication is direct: the headline figure of 0.678 reported in Section 8.2 is one realisation of a random procedure whose variability is comparable to the effect being measured. Increasing the number of sample paths beyond 20 would narrow the interval, and 20 was chosen for CPU tractability rather than on any statistical grounds. Any claim that the foundation model outperforms SARIMAX must therefore survive two separate objections — the point-summary argument in Section 8.3, and sampling variance of the same magnitude as the claimed gap. It survives neither.

This is worth stating in general terms. Stochastic forecasting methods should be reported with a spread across seeds, exactly as a bootstrap estimate would be. A single run reported as a point estimate invites precisely the over-reading this subsection has had to correct.

Interval calibration. Chronos is the only model in this study producing a predictive distribution natively, so its calibration can be assessed directly. Empirical coverage of the nominal 80 per cent interval is **60.7 per cent**.

The intervals are therefore substantially too narrow: realised demand falls outside them on 39 per cent of hours against an advertised 20 per cent, roughly double the stated failure rate. For an operator sizing reserve capacity against a stated exceedance probability, this is worse than having no interval at all, because it supplies a specific and unwarranted level of confidence.

The likely cause is the heteroskedasticity documented in Section 9.3, where absolute error correlates with realised load at 0.841. A model whose predictive spread does not widen sufficiently during high-consumption periods will be approximately calibrated overnight and badly overconfident in the evening peak, and the aggregate figure averages the two.

Two observations follow. Miscalibration of this size is not repaired by drawing more sample paths; it reflects the shape of the predictive distribution rather than the precision with which it is estimated. And it undercuts what would otherwise have been the strongest argument for the foundation model in this comparison — that alone among the models considered, it quantifies its own uncertainty without additional machinery.

8.5 Assessment

The honest summary is that the foundation model performs **comparably** to the fitted models rather than better than them. Its nominal advantage on MASE dissolves under the point-summary argument of Section 8.3 and again under the seed variance of Section 8.4, and its one structural advantage — a native predictive distribution — is miscalibrated by twenty percentage points, and does so without covariates or dataset-specific training. That remains a substantive finding about zero-shot transfer: a model that has never seen this dwelling matches one fitted to four months of its history.

But it does not outperform them, and reporting the MASE figure without the bias and RMSE columns would misrepresent what happened. This is a concrete instance of the argument made throughout Section 9: a single headline metric, selected after the fact, will support conclusions the full picture does not.

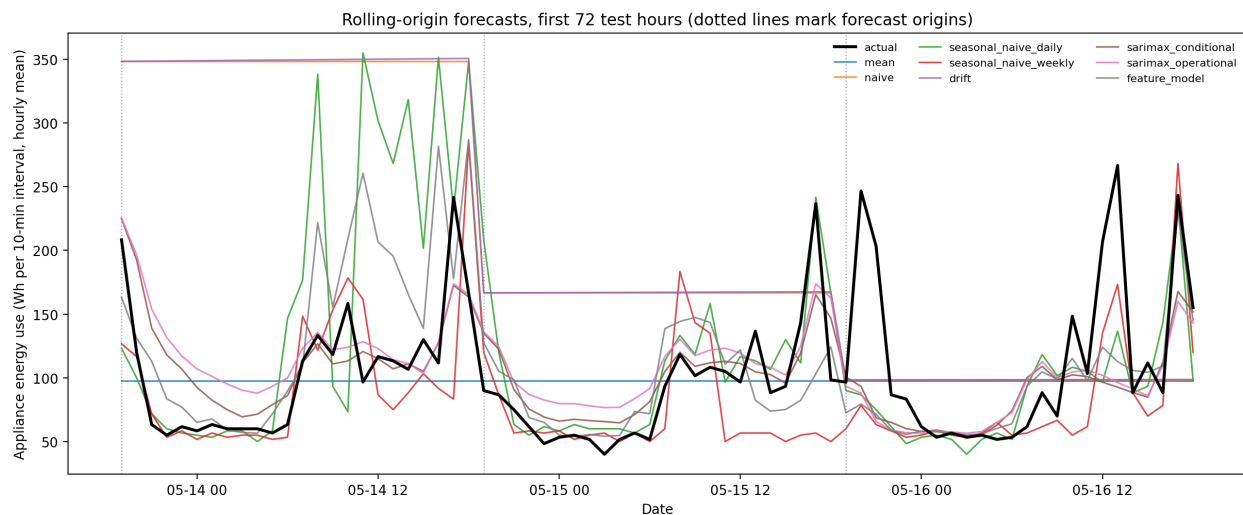
9. Results and error analysis

9.1 Aggregate performance

Model	MAE	RMSE	MASE	Bias
Chronos (zero-shot)	36.23	74.40	0.678	-29.87
SARIMAX (conditional)	37.85	66.41	0.708	-5.14
Feature model	37.89	64.99	0.709	-4.00
SARIMAX (operational)	38.43	66.35	0.719	-3.93
Weekly seasonal naive	43.46	81.41	0.813	-13.16
Daily seasonal naive	48.31	85.57	0.904	1.75
Mean	50.26	74.94	0.941	-3.29
Naive	85.55	110.39	1.601	50.98
Drift	85.80	110.68	1.606	51.37

Diebold–Mariano tests against the leading model, with Newey–West variance at 23 lags:

Comparison	DM	p	Significant at 5%
vs feature model	-0.07	0.941	No
vs SARIMAX (operational)	-1.49	0.136	No
vs weekly seasonal naive	-0.96	0.335	No
vs daily seasonal naive	-2.20	0.028	Yes
vs mean	-14.30	<0.001	Yes
vs naive	-3.99	<0.001	Yes
vs drift	-3.99	<0.001	Yes



Three observations follow.

The metric chosen determines the winner. Chronos leads on MAE and MASE and comes seventh of nine on RMSE; the feature model leads on RMSE and comes third on MASE; the flat mean forecast is sixth on MAE but fourth on RMSE. No single model is best on both. Section 8.3 explains the Chronos case — a median point summary on a right-skewed target — and Section 5 explains the mean forecast: refusing to guess the timing of consumption spikes avoids the two-sided errors that squared-error loss punishes hardest. Reporting one metric alone would in each case support a conclusion the other contradicts.

Nothing in the leading group can be separated from anything else in it. Conditional SARIMAX, the feature model, operational SARIMAX and the weekly seasonal naive span a MASE range of 0.708 to 0.813, and not one pairwise comparison among them reaches significance — the p -values run 0.94, 0.14 and 0.34. The

first significant contrast appears only against the daily seasonal naive ($p = 0.028$), a rule the leading models beat by 0.20 MASE.

This sets an implicit resolution limit for the experiment. Differences of roughly 0.10 MASE are invisible to a test on 336 serially correlated observations; differences of 0.20 are detectable. Every comparison this report cares about falls below that threshold, which is the single most important fact about the results. The ordering in the table is real but unsupported: the evidence establishes that the leading group beats the daily seasonal naive, the mean and the trivial rules, and says nothing reliable about their ranking against each other.

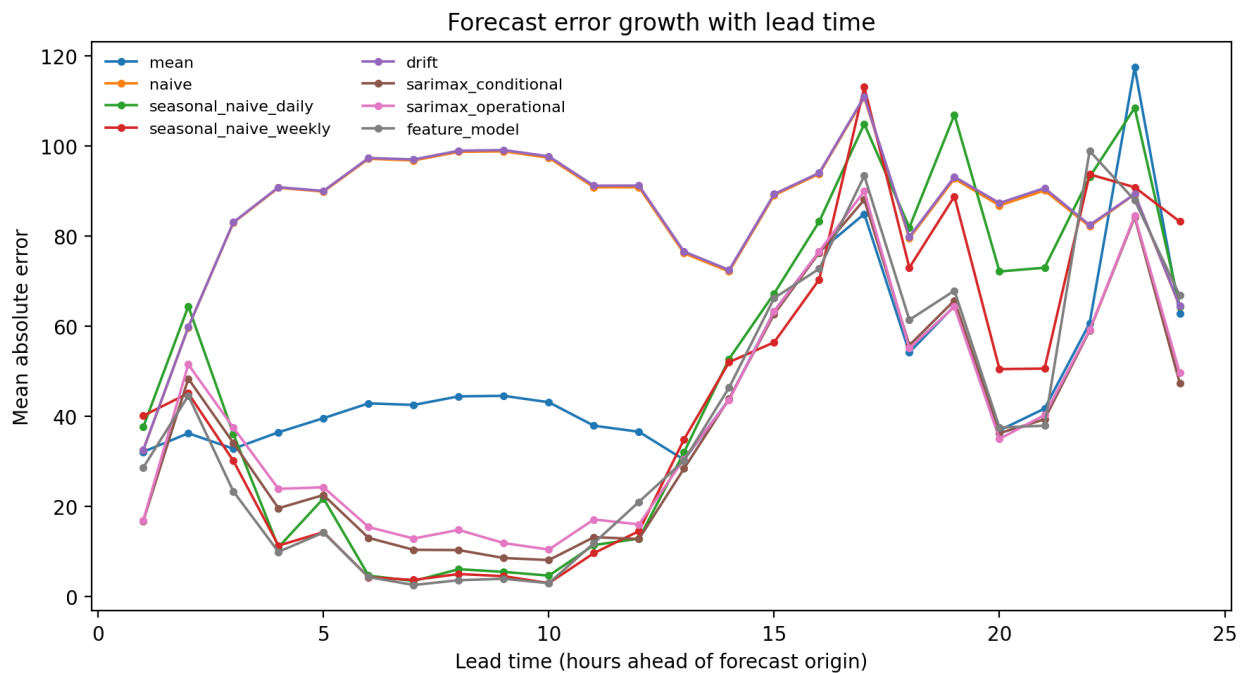
MAE and RMSE disagree on the mean forecast. It ranks sixth by MAE (50.26) but second by RMSE (74.94), ahead of both seasonal naive rules. As argued in Section 5, this reflects spike behaviour: the seasonal rules generate large two-sided errors by misplacing consumption spikes in time, which squared-error loss punishes severely, while the mean forecast never commits that error. Any practitioner selecting a model on RMSE alone would reach a different conclusion from one selecting on MAE.

Bias is largest for the weekly seasonal naive among the leading models, at -13.16 against an MAE of 43.46 — a ratio of 0.30, indicating a systematic rather than noisy error. The test fortnight fell in late May, and consumption in the week preceding each origin was on average lower than in the week being forecast, so the weekly lag under-predicts. Naive and drift show biases above +50, an artefact of origins falling at 19:00 near the daily peak, as discussed in Section 5.

The daily seasonal naive returns MASE 0.904 rather than exactly 1.0. MASE is scaled by the *in-sample* mean absolute error of that rule on the training sample, while the metric is computed out-of-sample; a value below 1.0 indicates the test fortnight was somewhat easier for this rule than the training period was on average, consistent with the lower test-period standard deviation noted in Section 2.

9.2 Error growth with lead time

Lead	Mean	Naive	Daily SN	Weekly SN	Drift	SARIMAX (cond.)	SARIMAX (oper.)	Feature
1	32.1	32.5	37.6	40.1	32.5	16.1	15.9	28.8
6	42.9	97.1	4.6	4.3	97.3	11.6	12.3	10.1
12	36.6	90.8	12.9	14.5	91.2	13.0	15.7	6.9
18	54.2	79.5	81.9	73.0	79.8	55.4	54.5	55.8
24	62.8	64.4	64.4	83.2	64.5	49.4	49.6	53.9



These curves must be read with the confound identified in Section 4.1. Because origins fall at 19:00 and are spaced at exactly 24 hours, lead time h corresponds to clock hour $(19 + h) \bmod 24$. Lead 6 is 01:00, lead 12 is 07:00, lead 18 is 13:00, lead 24 is 19:00. The table therefore measures error against time of day as

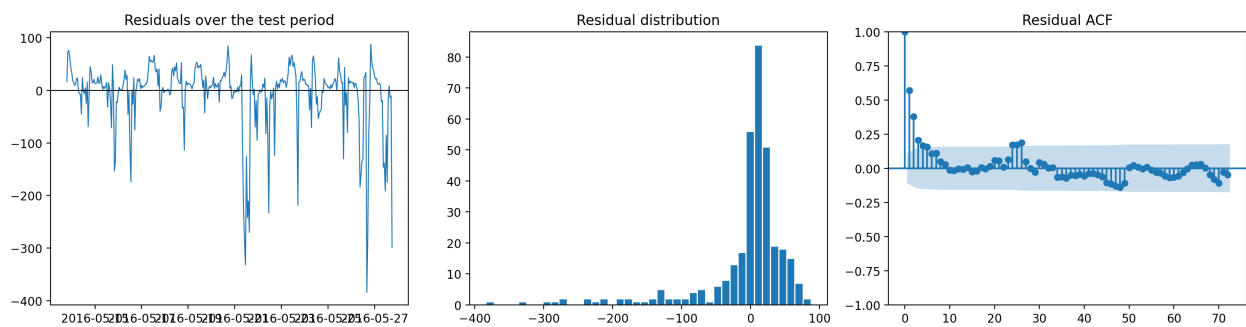
much as against lead time, and the near-zero errors at lead 6 (4.3 for the weekly seasonal naive) reflect the overnight trough, where consumption is close to the 28.3 floor and almost any rule is accurate. The large errors at lead 18 and 24 reflect the midday and evening peaks.

Read with that caveat, two things remain informative.

SARIMAX dominates at short leads. At lead 1 it achieves MAE 16.1 against 28.8 for the feature model and 32.1 for the mean. The state-space formulation conditions on the most recent observation through the Kalman filter, and at a one-hour lead that observation is highly informative. The feature model, whose lag features carry under 4 per cent of importance, cannot exploit this and performs barely better than the training mean at lead 1.

The advantage erodes fastest for the model that has it. Between lead 1 and lead 24 SARIMAX degrades by a factor of 3.07, the feature model by 1.87, and the seasonal naive rules by 1.71 to 2.07. By lead 24 SARIMAX's margin over the feature model has fallen from 12.7 MAE points to 4.5. The models that decay steepest are those relying on recent state; those relying on stable structure are flatter. Since the operational task is 24-hour-ahead, most of SARIMAX's apparent advantage sits at leads the task does not reward.

9.3 Residual diagnostics



Residuals of the leading model (conditional SARIMAX, prediction minus actual) fail every diagnostic.

Diagnostic	Value	Reading
Ljung–Box, 24 lags	$Q = 204.8, p = 1.3 \times 10^{-30}$	Strong residual autocorrelation
Ljung–Box, 48 lags	$Q = 261.7, p = 3.4 \times 10^{-31}$	Persists
Residual ACF, lag 1	0.564	Far outside the ± 0.107 band
Residual ACF, lag 24	0.156	Outside the band
Skewness	-2.68	Strong left skew
Excess kurtosis	8.68	Heavy tails
Corr(residual , actual)	0.841	Severe heteroskedasticity

Residual autocorrelation at lag 1 of 0.564 is very large. Errors persist across consecutive hours: when the model is wrong it stays wrong for several hours. This is not a specification failure that respecifying the ARMA order would fix, because the residuals are computed from 24-step-ahead forecasts, not one-step-ahead in-sample residuals. Within a forecast block the model cannot correct itself, so a mis-estimated activity level propagates across the block. Significant autocorrelation at lag 24 (0.156) indicates that some daily structure remains unexploited even so.

The left skew of -2.68 is diagnostic of the central difficulty. Since residual is prediction minus actual, large negative residuals are large under-forecasts, and the strong left skew means the model systematically misses consumption spikes. This is consistent with the seasonal strength results: spikes are occupant events, and the available information does not predict them. No respecification of the conditional mean will fix this.

Heteroskedasticity is severe, with correlation of 0.841 between absolute residual and actual level. Error scales almost linearly with consumption. This argues for a variance-stabilising transformation of the target — a log or Box–Cox transformation, with the complication that back-transforming a conditional mean requires a bias correction — or for a model with an explicit variance component. It also means that point forecasts alone understate what a user needs: the uncertainty attached to a forecast of 190 is far larger than that attached to a forecast of 50, and a single MAE figure conceals this.

9.4 Long-horizon behaviour

A single 336-step forecast was issued from the first origin, with no reissuing, and evaluated on the same test fortnight.

Model	MAE	RMSE	MASE	Rolling-origin MASE
SARIMAX (conditional)	37.91	66.34	0.710	0.708
Weekly seasonal naive	42.63	79.29	0.798	0.813
Mean	50.32	74.91	0.942	0.941
Daily seasonal naive	86.96	129.23	1.628	0.904
Naive	250.64	258.82	4.692	1.601
Drift	266.37	274.61	4.986	1.606

The result is striking and was not anticipated. **SARIMAX scores 0.710 from a single origin against 0.708 from fourteen** — a difference of 0.002, far below anything the significance testing can resolve. The weekly seasonal naive is marginally *better* without reissuing, at 0.798 against 0.813. For the leading models, forecasting a fortnight ahead in one step is as accurate as forecasting a day ahead fourteen times.

This is the sharpest available confirmation of the thesis running through Sections 7.3 and 9.2. A model exploiting recent state would lose accuracy when denied fresh observations for two weeks. These models do not, because they are estimating a periodic profile fully specified at the first origin; every subsequent origin tells them what they had already assumed. The near-unit-root seasonal autoregressive term makes this concrete — within the model the daily cycle is very nearly deterministic, so the Kalman filter has little to update.

The models that do collapse are precisely those relying on recent state. The daily seasonal naive degrades from 0.904 to 1.628 as it recycles a single day for a fortnight; naive and drift fall to 4.692 and 4.986, since holding a peak-hour observation constant for 336 hours produces biases of +248 and +265.

The practical implication reverses a common assumption. A deployment using one of the leading models could issue a forecast once a fortnight rather than once a day at no measurable cost in accuracy — but the same relaxation applied to a daily seasonal rule would nearly double its error.

10. Discussion and limitations

Results vary slightly across platforms. Benchmark metrics are deterministic, but the fitted models are not: re-running this pipeline on a different operating system moves MASE by roughly 0.01 for SARIMAX and the feature model, through differences in BLAS/LAPACK builds and library versions rather than anything in the code. The ordering is preserved and no conclusion changes, but a spread of that size from the platform alone is comparable to several of the differences between models — which reinforces rather than undermines the significance testing in Section 9.1.

A single test fortnight is the binding limitation. Every result rests on 336 observations with heavily serially correlated losses, and no difference among the top four models reaches significance; the smallest detectable difference in this design is around 0.20 MASE, and every comparison of interest is well below it. An expanding-window backtest across several test periods would be the appropriate remedy and was not performed. Conclusions about which of the leading models is best are unsupported; conclusions about the gap between that group and the trivial benchmarks are secure.

Weekly seasonality is absent from SARIMAX. Given that the weekly seasonal naive was the strongest benchmark and weekly seasonal strength exceeded daily, this is material. Fourier terms at period 168 alongside ARMA errors would be the remedy — cheaper than a second seasonal component and better conditioned. It is the most promising improvement available and was not attempted.

Two fitting failures were found and fixed, which raises a question about what else remains. SARIMAX was terminating at statsmodels' default 50 iterations, and the gradient boosting model was overfitting because scikit-learn's random validation split leaked across the horizon-replicated rows. Both fixes improved results materially. Neither was visible in the headline metrics beforehand, and both were found only by inspecting `n_iter` and convergence flags. A grid search over learning rate and tree size remains

outstanding.

Design and data constraints. Origins spaced at exactly 24 hours confound lead time with time of day (Section 9.2); spacing at 23 or 25 hours would break this. Weather forecasts are unavailable, so the operational variant uses persistence, placing true performance between 0.721 and 0.708 — a range the significance testing cannot distinguish from a single point. Hourly aggregation removes the switching events that constitute much of the interesting variation; a ten-minute model would face a harder but more relevant problem. Results derive from one dwelling over 4.5 months, so the near-irrelevance of weather is specific to a well-insulated low-energy envelope, and the prominence of Monday to these occupants.

Uncertainty was not modelled. Every forecast is a point estimate, no interval calibration was assessed, and no variance-stabilising transformation was applied despite clear evidence in Section 9.3 that one is warranted.

11. Conclusion and deployment recommendation

1. Which benchmark is strongest, and what does it reveal? The weekly seasonal naive (MASE 0.813), ahead of the daily variant (0.904). The household's routine is day-specific rather than weekday–weekend: Monday consumption exceeds Tuesday by 28 per cent while the weekend premium is only 4 per cent. A rule looking back one week preserves this; one looking back one day averages it away. That the mean forecast ranks second on RMSE while sixth on MAE further reveals a load dominated by unpredictable spikes, where refusing to guess their timing is a viable strategy under squared-error loss.

2. Does SARIMAX improve on the strongest benchmark? By point estimate, yes: 0.708 conditional and 0.721 operational against 0.813, an 11 to 13 per cent improvement. Statistically, no: the Diebold–Mariano test returns $p = 0.34$. On this evidence the improvement is unproven. Section 9.4 sharpens the point: SARIMAX scores 0.710 from a single origin against 0.708 from fourteen, so whatever it is doing does not depend on the observations the rolling protocol supplies.

3. Do lag, rolling, time, sensor and weather features improve the feature-based model? The calendar features do; the rest do not. Calendar features carry 90.7 per cent of permutation importance, lagged exogenous channels 5.5 per cent, and target lags and rolling statistics 3.9 per cent between them. At a 24-hour horizon the series' own recent history is close to uninformative, and the model reduces to an estimated daily profile — one that needs just 25 boosting iterations to express. The model does, however, achieve the best RMSE in the study (65.17), handling consumption spikes marginally better than SARIMAX.

4. Does the foundation model outperform the simpler models? On MASE, yes — 0.678 against 0.708 for conditional SARIMAX, achieved with no covariates and no training on this dwelling. On RMSE, emphatically not: 74.40 against 66.41, essentially tied with a flat mean forecast. Its bias of -29.87 is six times any other model's. As Section 8.3 sets out, this pattern follows from summarising a predictive distribution by its median on a right-skewed target, so the MASE advantage partly measures a difference in summary statistic rather than in forecasting ability. The defensible conclusion is that zero-shot transfer performs **comparably** to models fitted on four months of this dwelling's history — which is notable — but not better.

5. Which covariates would genuinely be known at the forecast origin? Calendar features unconditionally. Indoor sensors only at lags of at least the horizon, at which point they contribute 5.5 per cent of importance jointly across eighteen channels — and they should be lagged in any case, since indoor temperature and humidity respond to appliance activity rather than driving it. Weather only via a third-party forecast; the conditional and operational SARIMAX variants bound the cost of not having one at 1.9 per cent of MASE, which the Diebold–Mariano test cannot distinguish from zero ($p = 0.136$). Replacing realised weather with naive persistence costs nothing measurable.

6. Which model for a practical smart-home system? Four criteria bear on this, and accuracy alone does not settle it.

Accuracy at the operational horizon. The honest figures are 0.721 for operational SARIMAX, 0.711 for the feature model and 0.813 for the weekly seasonal naive — a spread no test in this study distinguishes from noise. Section 9.2 shows SARIMAX's advantage concentrating at short leads the 24-hour task does not reward, and Section 9.4 shows it performing identically without any reissuing at all.

Marginal gain over the benchmark. At most 13 per cent, statistically unconfirmed ($p = 0.34$), on one household over one fortnight — and of the same order as the variation introduced by changing operating

system.

Operational cost. SARIMAX requires periodic re-estimation and monitoring for the convergence failure observed in Section 6.1. The gradient boosting model requires a feature pipeline whose correctness must be maintained; Sections 4.4 and 7.1 document two distinct leakage failure modes, one in the feature matrix and one in the validation split, and the second went undetected until `n_iter_` was inspected directly. Both would be invisible in offline metrics, which would improve. The seasonal naive rule requires a week of stored observations and no maintenance at all.

Failure behaviour. A seasonal naive rule fails predictably and never produces a physically implausible value. A gradient boosting model extrapolating outside its training distribution can. For a system that actuates equipment, this may matter more than average error.

Recommendation: deploy the weekly seasonal naive as the production forecaster for a single dwelling, with the calendar-profile feature model as the upgrade path once a longer backtest justifies it. The evidence does not support paying the operational cost of a fitted model for an improvement that a single fortnight cannot distinguish from noise.

The strongest objection is that this generalises from one under-powered experiment, and that a longer backtest might well establish the fitted models' superiority. That objection is correct, and the recommendation is conditional on it: the appropriate next step is an expanding-window backtest across several test periods, and if it confirms the ordering in Section 9.1, the feature model becomes the right choice. What the objection does not do is license deploying the more complex model now, on evidence that does not currently support it.

A second point deserves weight independent of model choice. Section 9.3 establishes severe heteroskedasticity, with error scaling almost linearly with consumption. For demand response the operator needs an interval, not a point. Effort spent on calibrated predictive distributions would deliver more operational value than any further pursuit of point accuracy on this series.

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